

LESSON PLAN

SUBJECT	Physical Science	DATE	09/09/2026
GRADE	11	TIME	09:00
SUBJECT TOPIC (BROAD)			
Electricity and magnetism			
LESSON TOPIC			
Electrostatics			
ASSESSMENT STANDARDS			
By the end of this lesson, learners must be able to:			
• State Coulomb's law in words • Calculate using the formula $F = \frac{kQ_1Q_2}{r^2}$ for charges in one dimension (1D) restricted for 3 charges and for charges in two dimension.			
PRIOR KNOWLEDGE			
• Positive and negative charges • Electrostatic force • Conservation of charge • Vectors and scalars			
Teach STRATEGIES			
Learner centred teaching			
Investigative learning			
Direct teaching			
TEACHING AIDS			
Projector: used to display multi-charge diagrams, problem statements and step by step vector resolution layouts to the whole class			
Physics education technology (PhET) Simulation: used to visually demonstrate how vary charges and distance alter force vectors in real time			
CAPS Aligned textbook: used as the primary source for structured 2D practice problems and homework tasks			
LEARNING AIDS			

PhET Simulation
CAPS Aligned Textbook

TIME INTERVAL	TEACHERS ACTIONS	LEARNERS ACTINS	ASSESSMENT TOOL
09:00 – 09:15	Administer baseline test and walks around evaluating learners	Learners individually write the test in 10 minutes and swap books to peer mark	Baseline test
09:15 – 09:25	Teacher puts ballons, wool cloth and paper bits on desk and ask: ‘in your groups, rub the ballon and make it pick up paper, then bring the two rubbed ballons together. What do you see and why does this happen? Where is the force coming from?’	Investigate the scenario: learners rub ballons, observe attraction to paper and repulsion between the two ballons. Groups discuss and demonstrate to class, ‘we charged the ballons by friction. Ballons became negative and induces positive on the paper bits and attracted each other. But two ballons together repel because they are both negative.	Practical observation and group discussion
09:25 - 09:35	Teacher projects simulation and ask 3 learners to come and test it. Beginning with q2 at 2μC r=4 Learner 1: Change Q1 from 1μC to 2μC. What happens to the force? Learner 2: Double the distance from 4 to 8cm, what happens? Learner 3; Half the distance? So what is the rule?	Learner 1 changes charge: ‘force doubled’ Learner 2 drags the distance, class measures ‘force became 4x smaller therefore F directly proportional to Q1Q2 and inversely proportional to the squared distance. Class together combine the formula $F = \frac{kQ_1Q_2}{r^2}$	Simulation and learner demonstration. Assess discovery of relationships
09:35 – 09:55	Give learners examples: Q1=2μC at 0m and Q2= -4μC at 0.3m. Ask learners to verify simulation values by calculation. Teachers assign class activity on one-dimension calculations and walks around to evaluate learners’ calculations and how they convert μC from the example.	Learners in pairs use a calculator to calculate the given examples: convert 2μC to $2 \times 10^{-6}C$. They use the formula to substitute the given values and get 0.8N as their final answer. Learners come t the board to set values in the PhET to confirm their final answers. Learners start doing their assigned class activity based on one-dimension calculations individually.	Classwork calculations and verification with PhET simulation
09:55 – 10:00	Who can state the law you discovered today without reading it? How can you calculate the force between	One learner states Coulomb’s law discovered from the simulation. All learners write summary of the law in their	Self assessment and verbal questioning

	charges? Teacher assigns homework	own words and start with their homework.	
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SIMULATION: Coulomb's Law - Macro Scale

Link: https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_en.html

<https://phet.colorado.edu/en/simulation/charges-and-fields>